

● Detecting Deepfakes in the Age of Generative AI

Noemi Roggero

Data Science Meetup Grenoble

26/02/2026

● Context: joint PhD thesis between Probayes and Inria (Nice)

Generalizable Deepfake Detection

STARS Team @ INRIA (Spatio-Temporal Activity Recognition of Social interactions):

Analysis of long-term human interactions in their natural environment. Focus on anomaly detection, object/activity tracking, video generation

Probayes @ La Poste:

AI experts from Computer Vision to LLMs. Use-case of deepfake detection: identity verification for Banque Postale



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C. Yahchouchi

OUTLINE●

01 Introduction

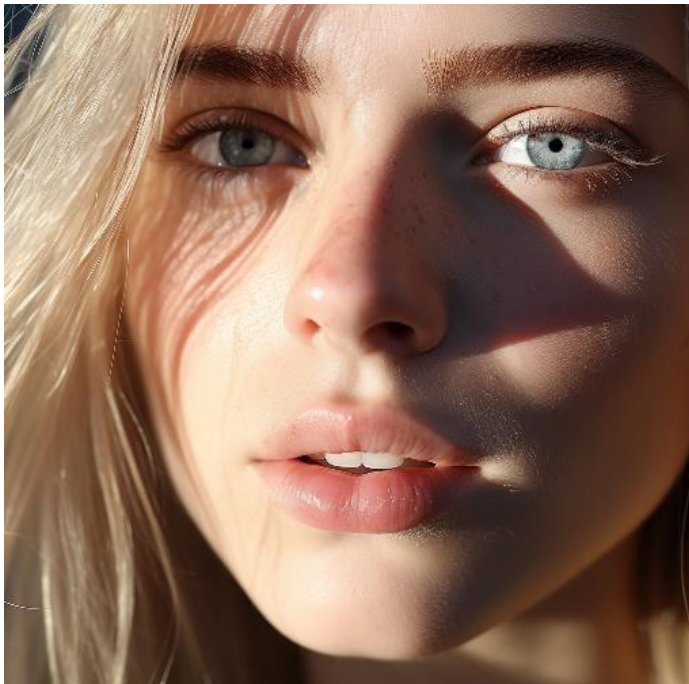
02 State of the art

03 First research results

04 Research directions

● What is a deepfake?

- **Deepfakes** are images, videos, or audio that have been edited or generated using artificial intelligence, AI-based tools or audio-video editing software.
- These videos or images can be fully synthesized, edited with AI, used to change the identity of the subject or to animate it



MidJourney6



MobileSwap

● Development of deepfakes since 2015



2015

Pre-Deep learning

Face2Face



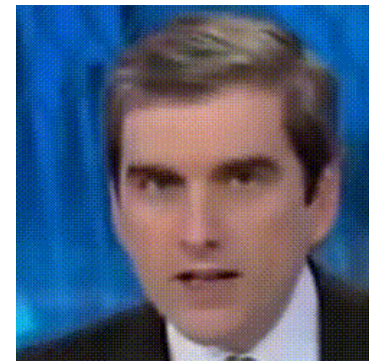
2018

DeepFakes



2019

GANs



2022

GANs



2025

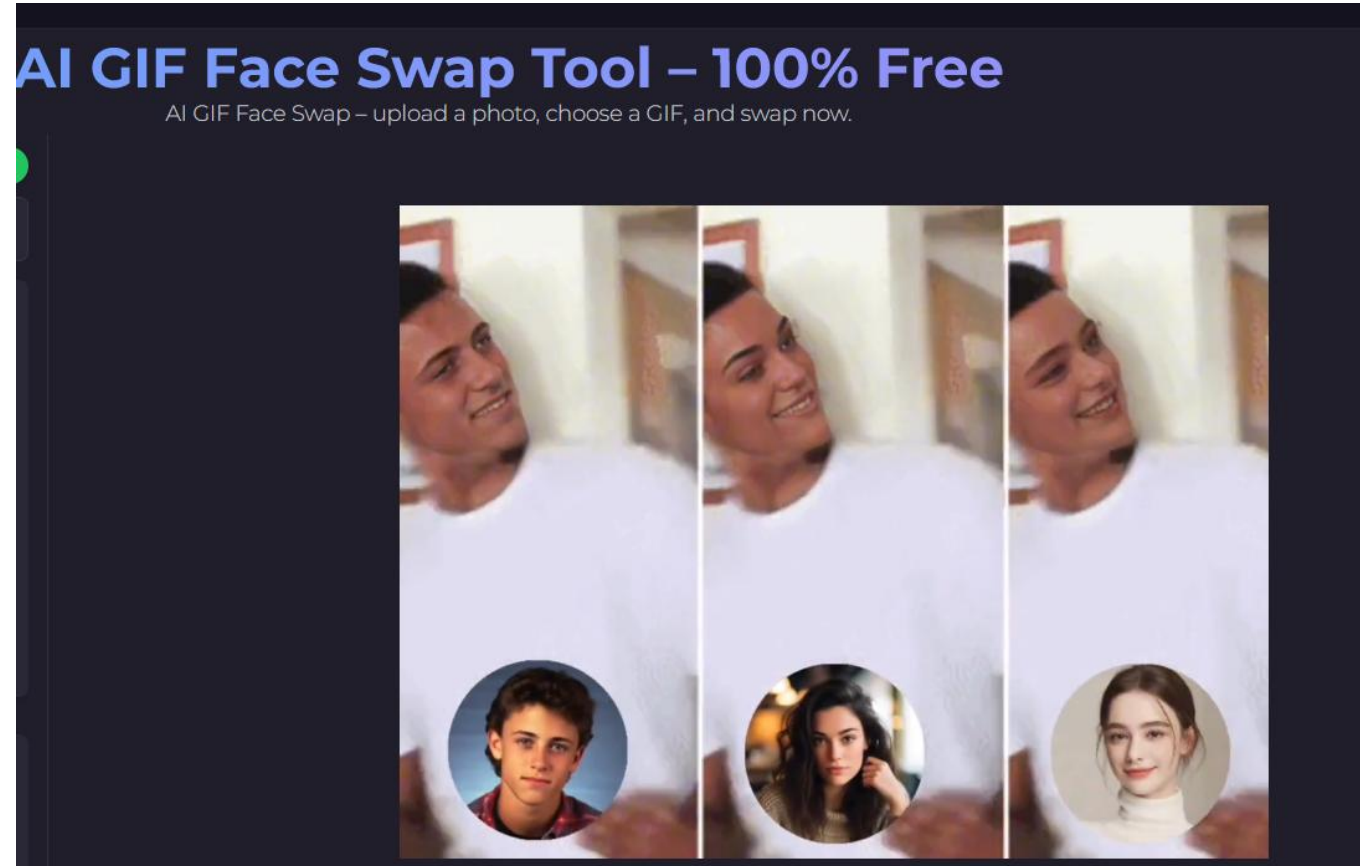
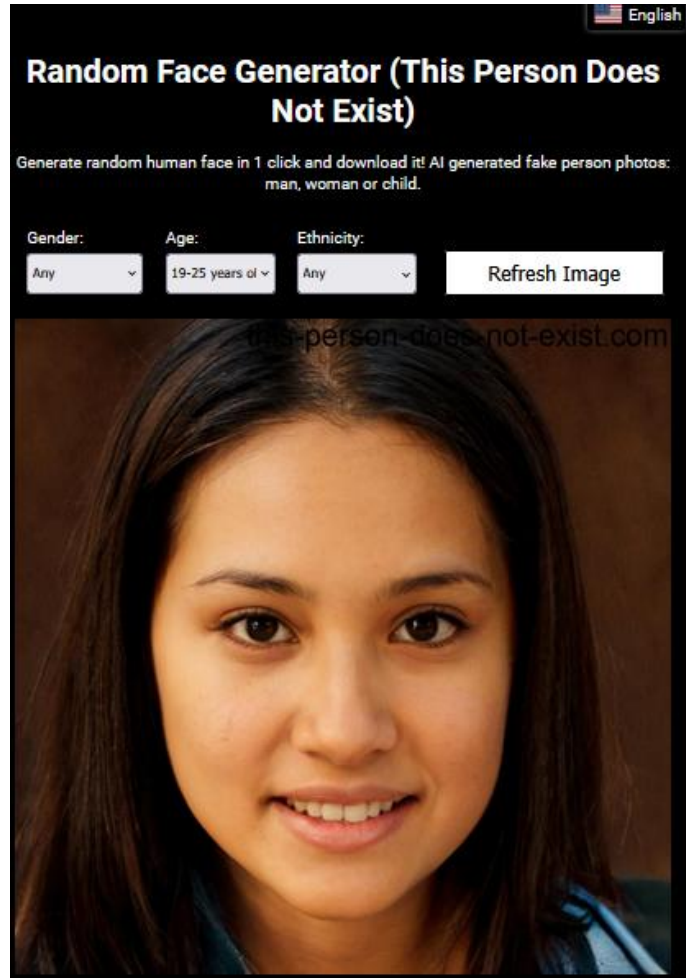
Diffusion models (Veo3)

● Deepfakes today



https://www.tiktok.com/@c.de.l.ia/video/7455746933496270102?is_from_webapp=1&sender_device=pc&web_id=7610835109243520534

● Can I play with deepfakes too?



● I play with deepfakes – Geek version

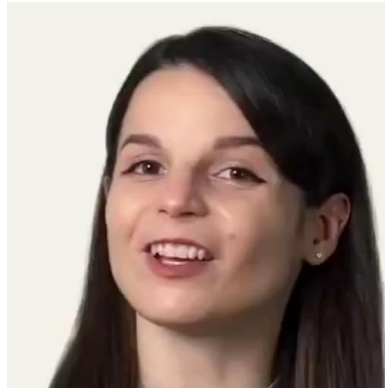
Spaces YaohuiW / LIA-X like 80 Running on ZERO Get PRO

App Files Community 4

LIA-X: Interpretable Latent Portrait Animator

Toward Interactive Portrait Animation and Editing

[\[Technical Report\]](#) | [\[Project Page\]](#) | [\[Code\]](#)



● Risks for our society

1

Fake news and manipulation

- Influence on geopolitics
- Chaos



● Risks for our society

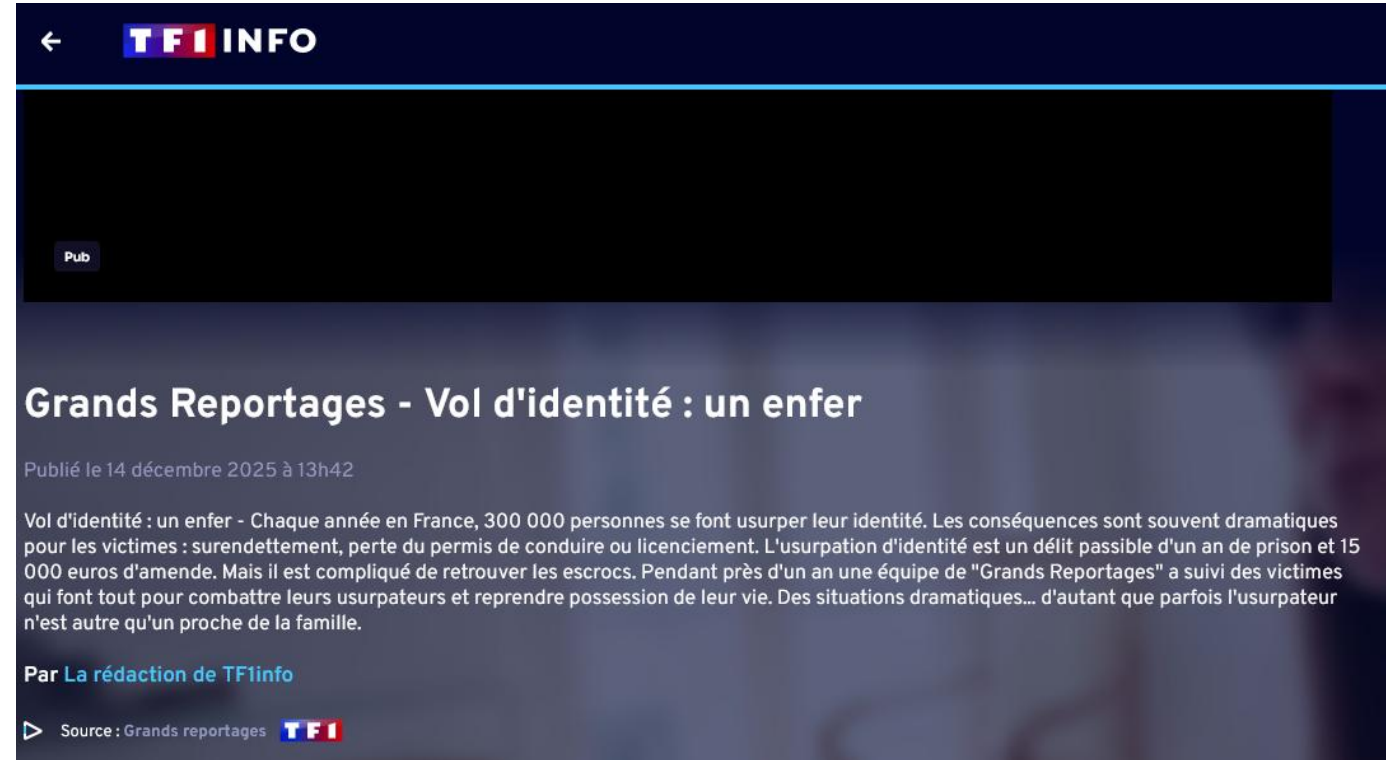
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Fake news and manipulation

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Identity theft



● Risks for our society

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Identity theft

- Challenge for facial authentication
- Fake (video) calls



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Identity theft

- Challenge for facial authentication
- Fake (video) calls
- Fake porn of real women and children

Grok is undressing women and children.
Don't expect the US to take action

Moira Donegan



Elon Musk's reckless and degrading AI could be built differently.
But Americans will have to speak up



📷 'The incident is a lesson in the dangers of rapid and unregulated technology: it is not a coincidence that among many users, the first thing they thought to do with AI was to harass and degrade women.' Photograph: Anna Barclay/Getty Images

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Cybersecurity risks

- Fake profiles

Instagram

Se connecter

S'inscrire

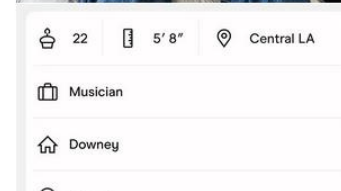


lilmiquela ✓

Miquela

1 462 publications 2,3 M followers 2 148 suivi(e)s

22 🍷 LA 🤖 Robot 🧠



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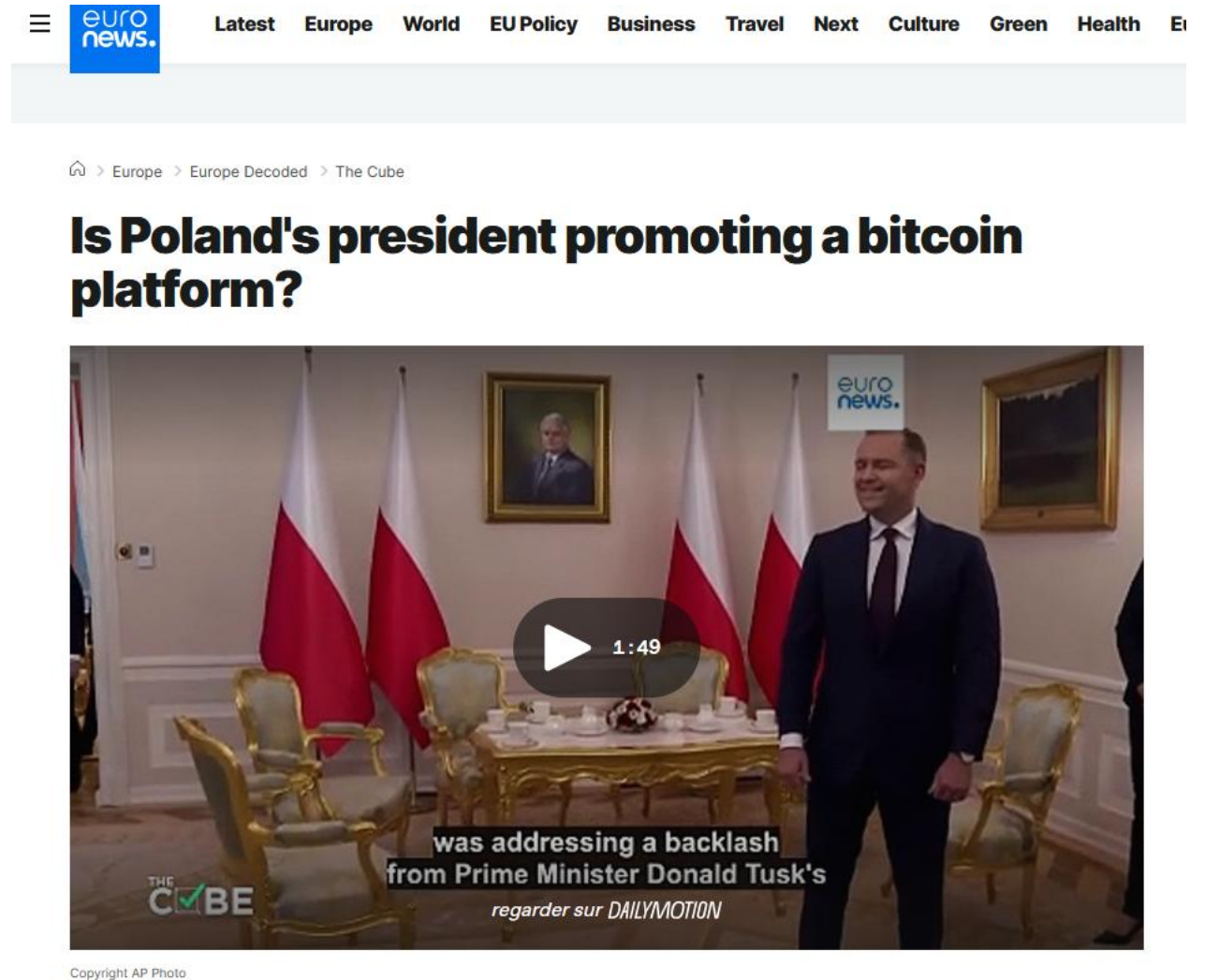
Identity theft

- Challenge for facial authentication
- Fake (video) calls
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Cybersecurity risks

- Fake profiles
- Large-scale scams



By [Tamsin Paternoster](#) & [Estelle Nilsson-Julien](#)

Published on 19/01/2026 - 16:10 GMT+1 • Updated 20:59

● What are different types of deepfakes ?

Face swap

We exchange two people's face in a real video

Models: FaceSwap, SimSwap, e4s, FaceDancer....

Face reenactment/Talking face

We animate an image by using a reference video or a set of landmarks

Models: SadTalker, Wav2Lip, Hyperreenact...

Entire face synthesis

We generate a completely fake image or video from scratch

Models: WhichIsReal, MidJourney6, StyleGAN...

Face editing

We edit certain parts of the image with a prompt

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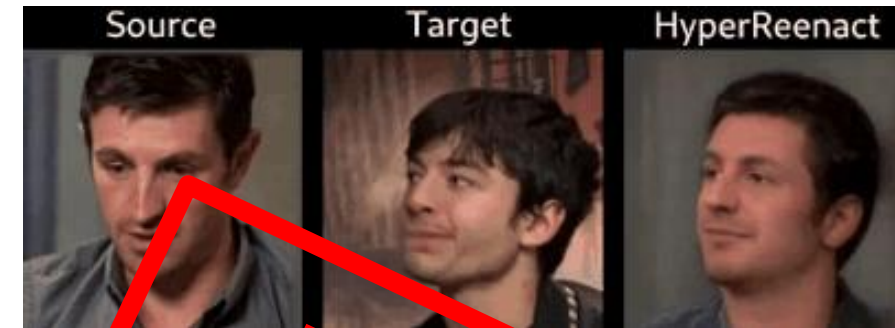
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WhichIsReal



MidJourney

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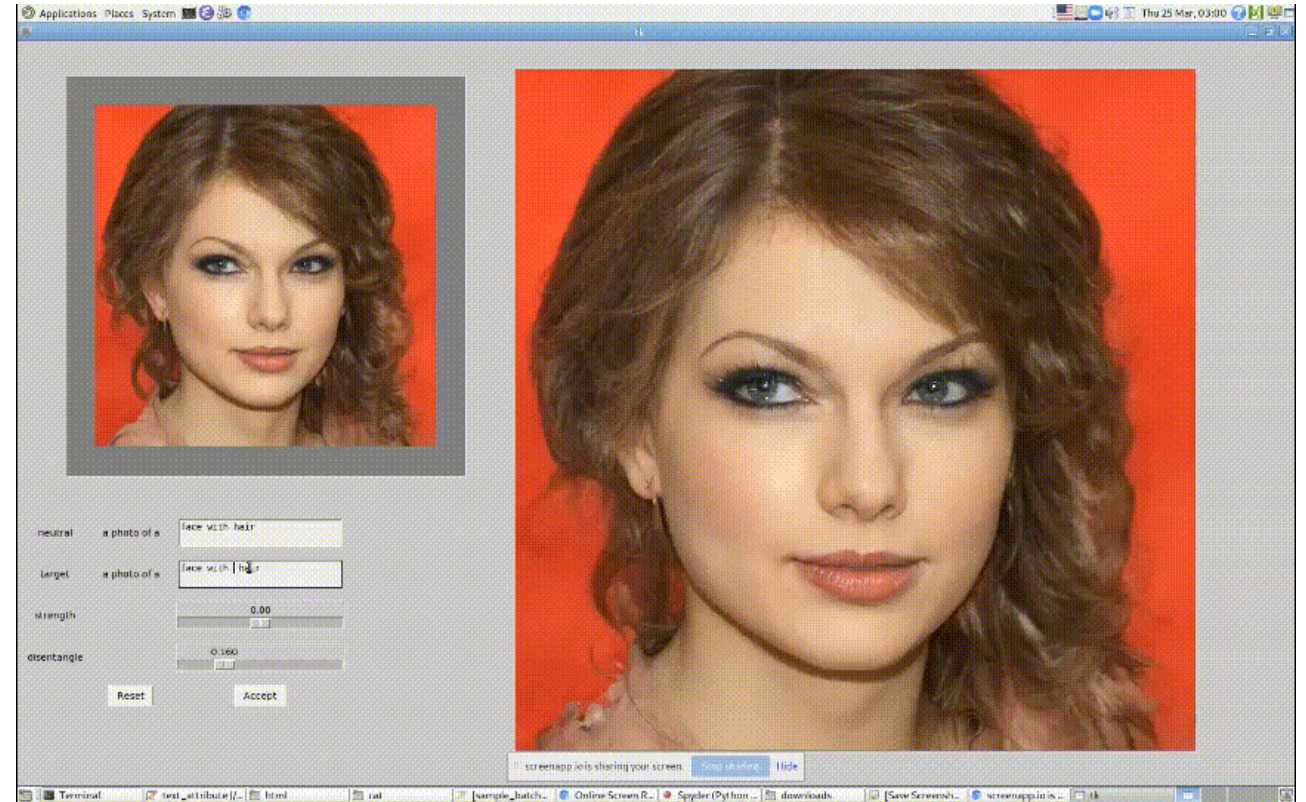
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StyleCLIP

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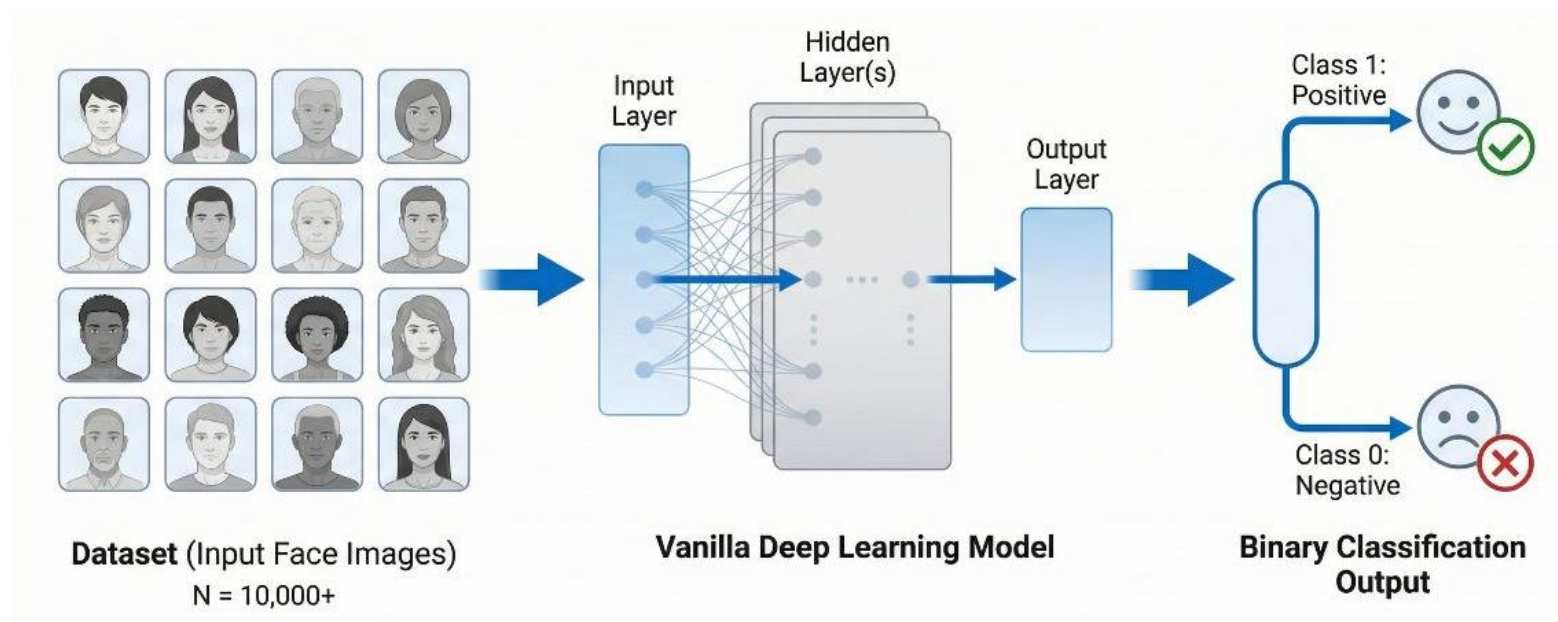
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● Struggle towards the generalization of deepfake detection



Dataset
- Which manipulations are included in the dataset? - SOTA benchmarks are outdated (2019) - Are there identity or dataset-specific biases?

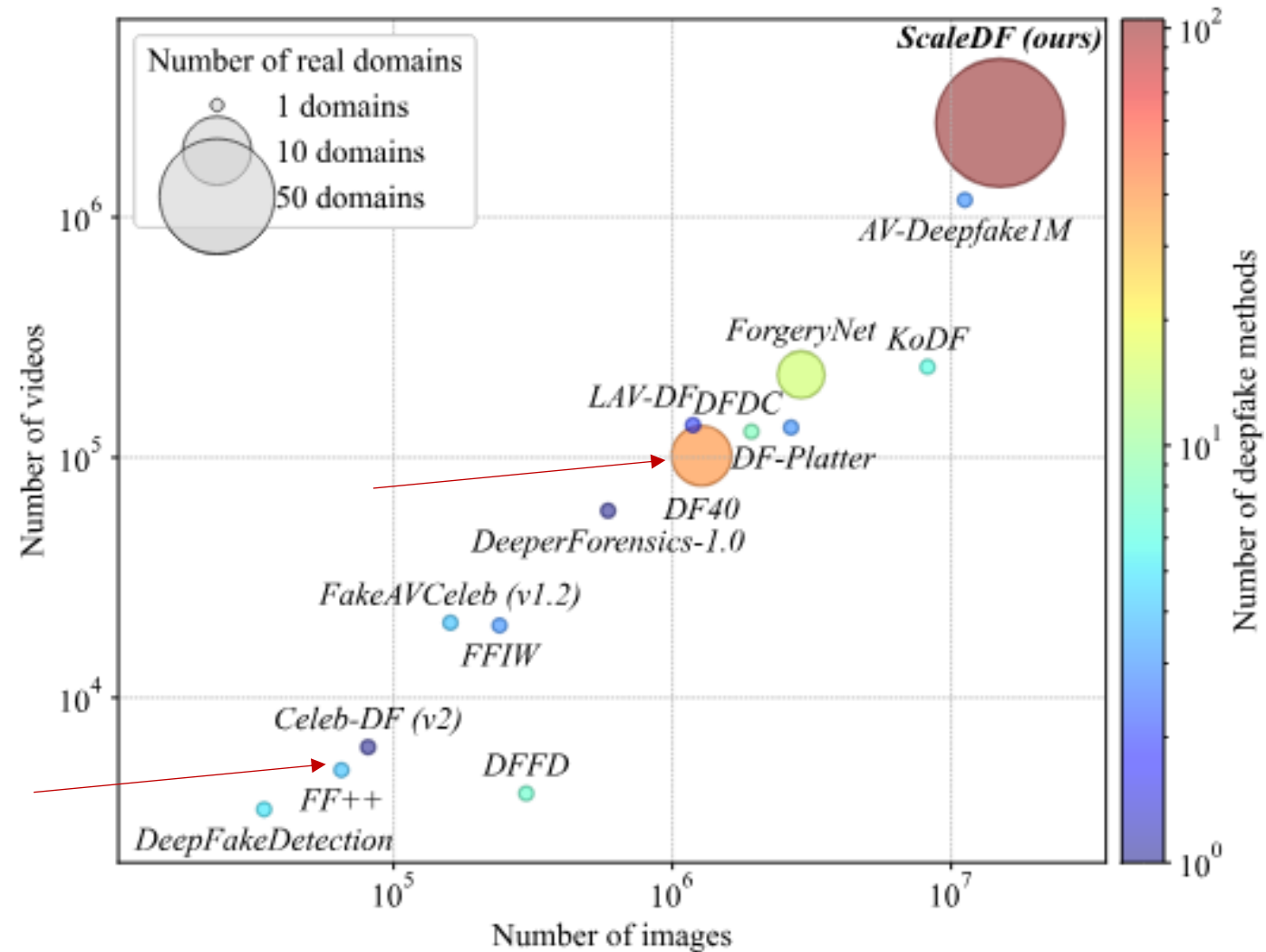
Model
- Which backbone is used? Spatial or temporal approach? - On which clues does it focus on? - Does it “understand” the different forges?

Objective
- Is binary classification the best objective? - If not, how do we define classes? Evaluation: no standard across papers

Dataset completeness: from FaceForensics++ to ScaleDF

Observed scaling **power law** as for LLMs:
We can reach $AUC = 0.95$ with 300 real domains or 700 deepfake methods!

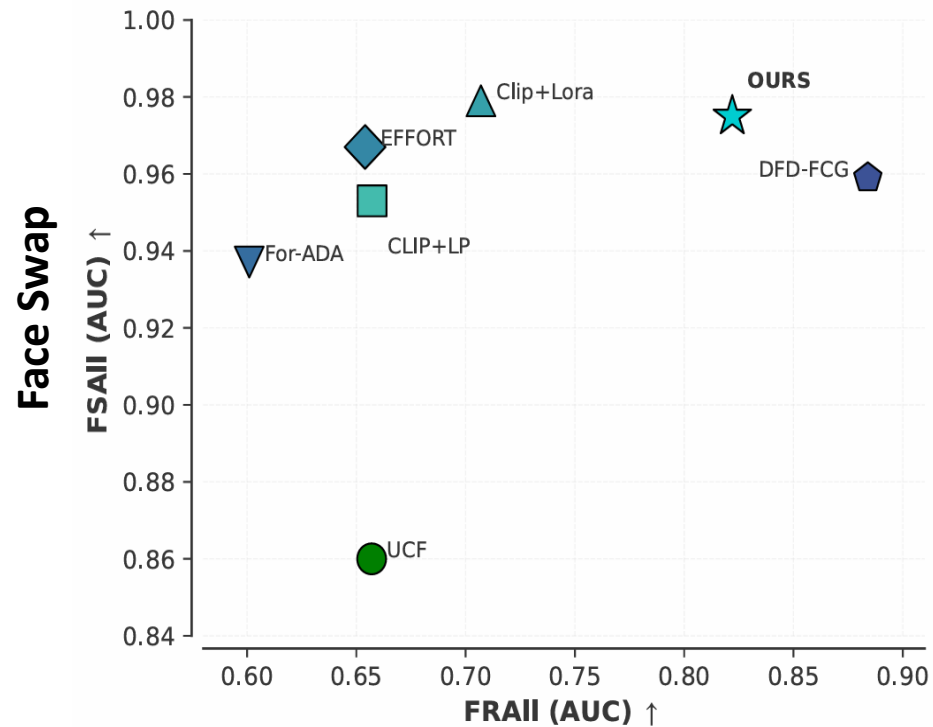
We are far from this limit



Wang et al. (2025)

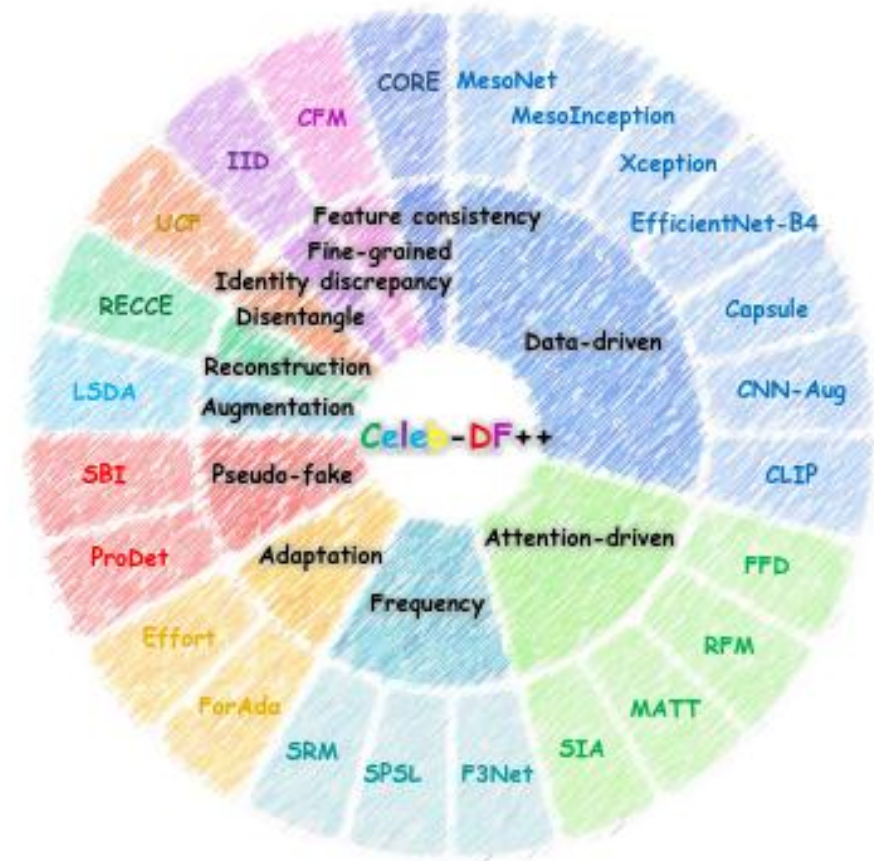
Models: SOTA efforts towards generalization

- Lack of **generalization** to out-of-distribution techniques due to
 - Manipulation-dependent cues
 - Identity bias
 - Learning of background or dataset-dependent patterns
 - Simplicity bias



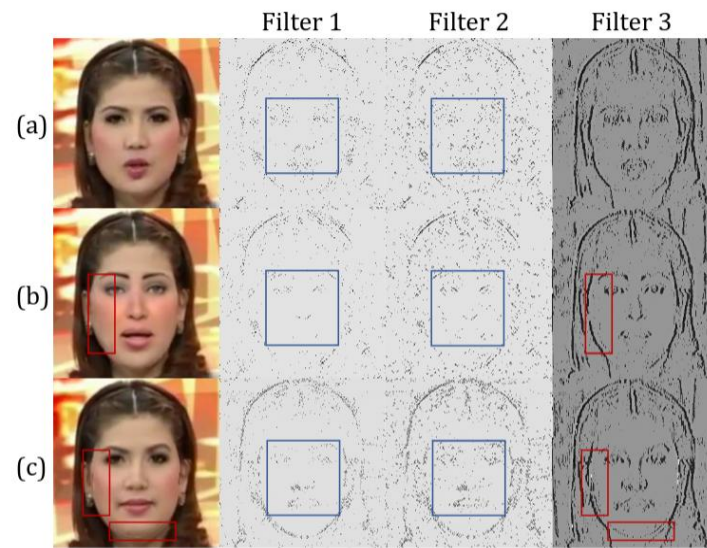
Face Reenactment

C. Yahchouchi

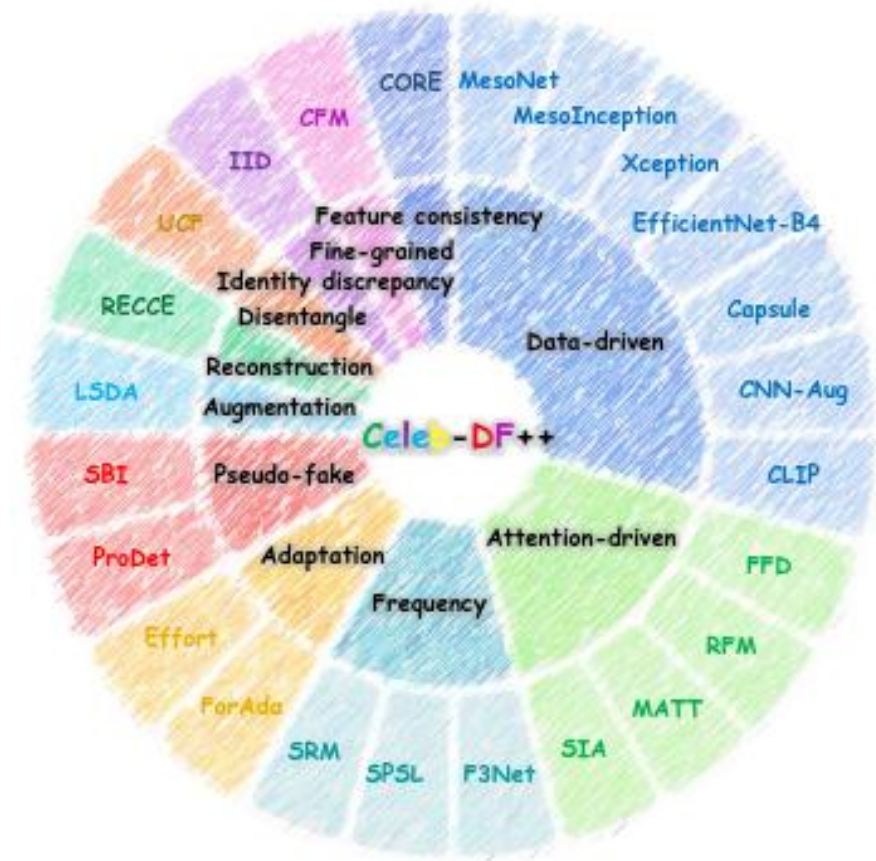


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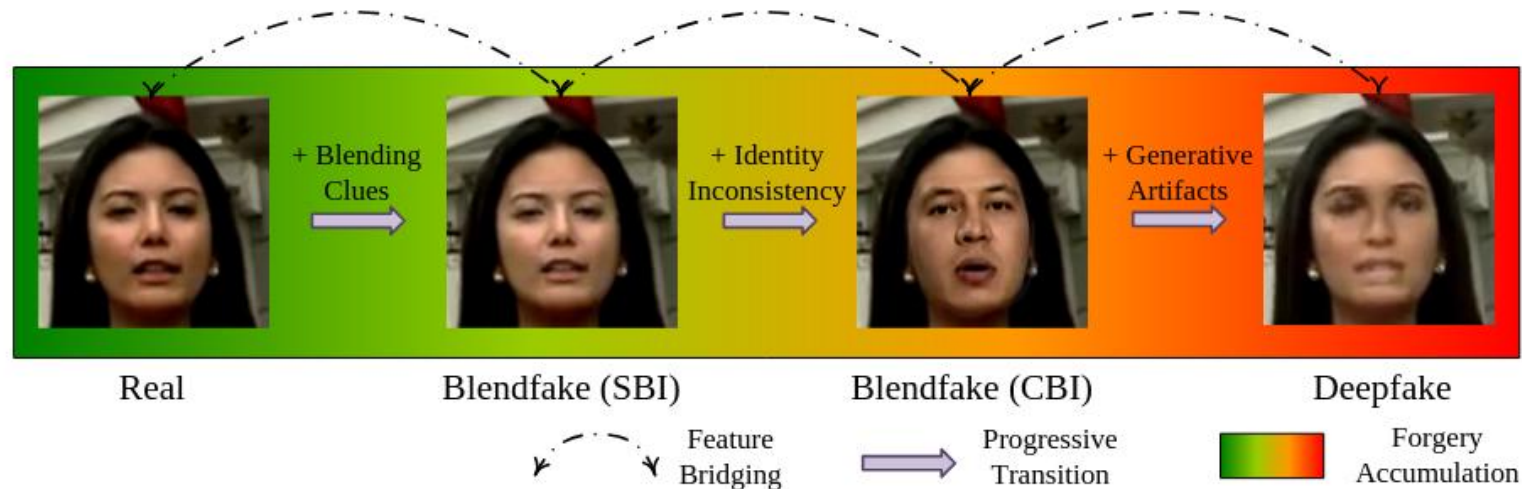


Luo et al. (2021)



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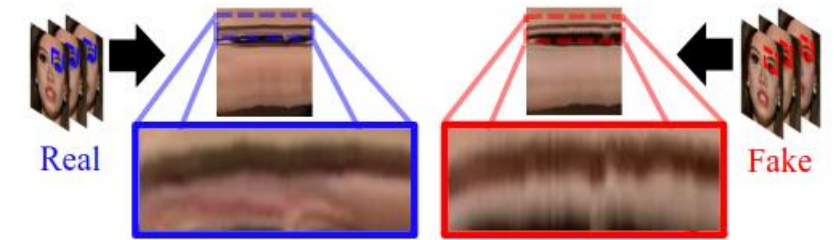
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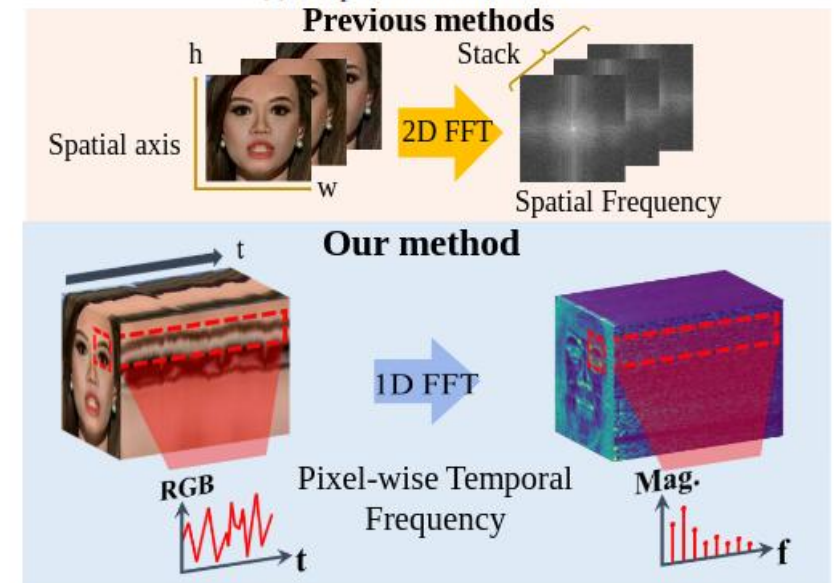
Chang et al. (2024)

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- Video-based approaches investigate **temporal** inconsistencies



(a) Temporal artifacts in Videos

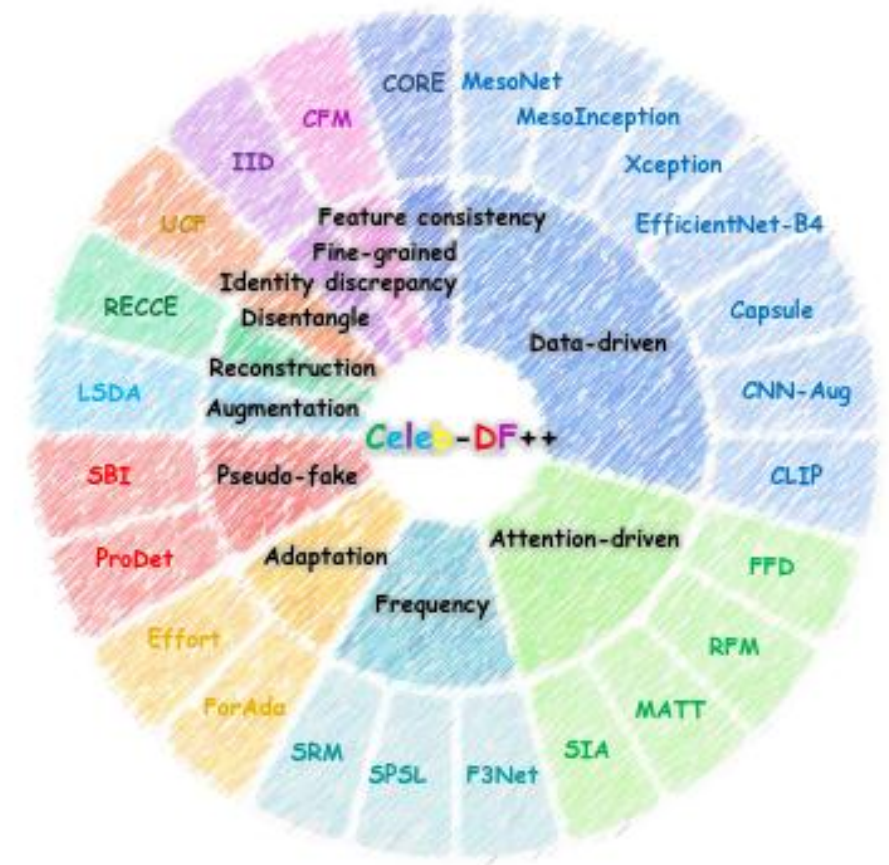


(b) Frequency extraction method

Kim et al. (2025)

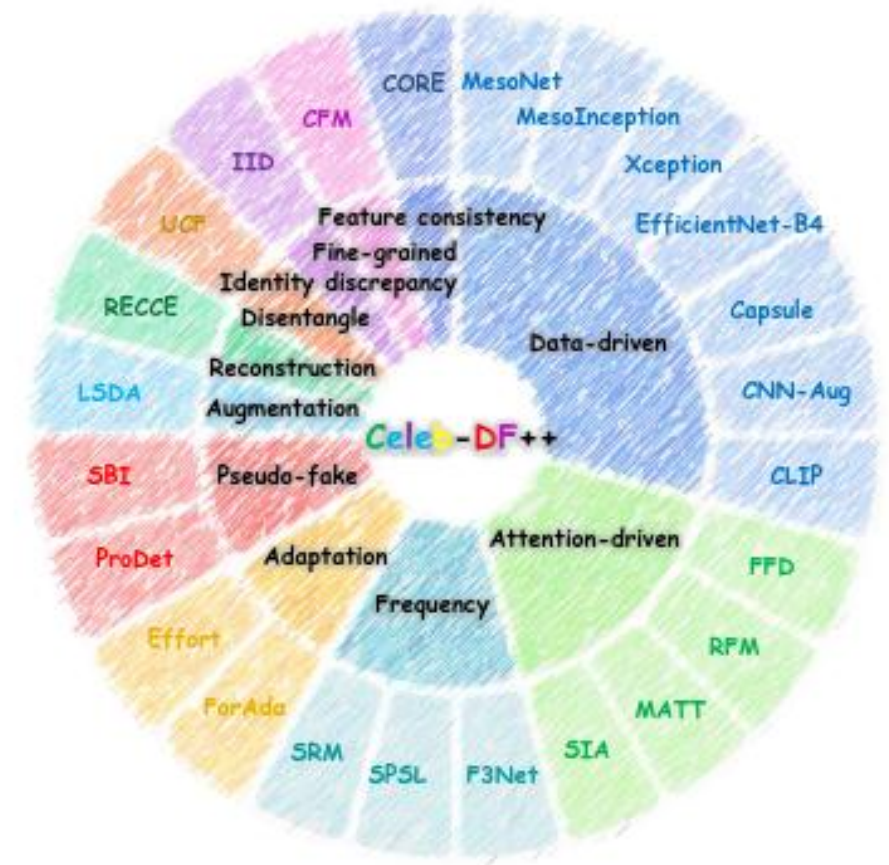
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- Mixture of **Experts** combine different specialized models into one



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- Video-based approaches investigate **temporal** inconsistencies
- Mixture of **Experts** combine different specialized models into one
- Many works use the frozen weights of a **foundation model** (models you don't want to train at home, e.g. CLIP) combined with trainable adapters



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Adapting CLIP to overcome the simplicity bias

Problems addressed:

- insufficient **benchmarking** in the literature
- learning of **irrelevant** features during training
- Amplification of **simplicity bias** under binary classification

Method:

- **Extensive** cross-dataset, cross-manipulation and out-of-distribution **evaluation**
- CLIP **adapters** + **SIFER** (feature-sieving mechanism)

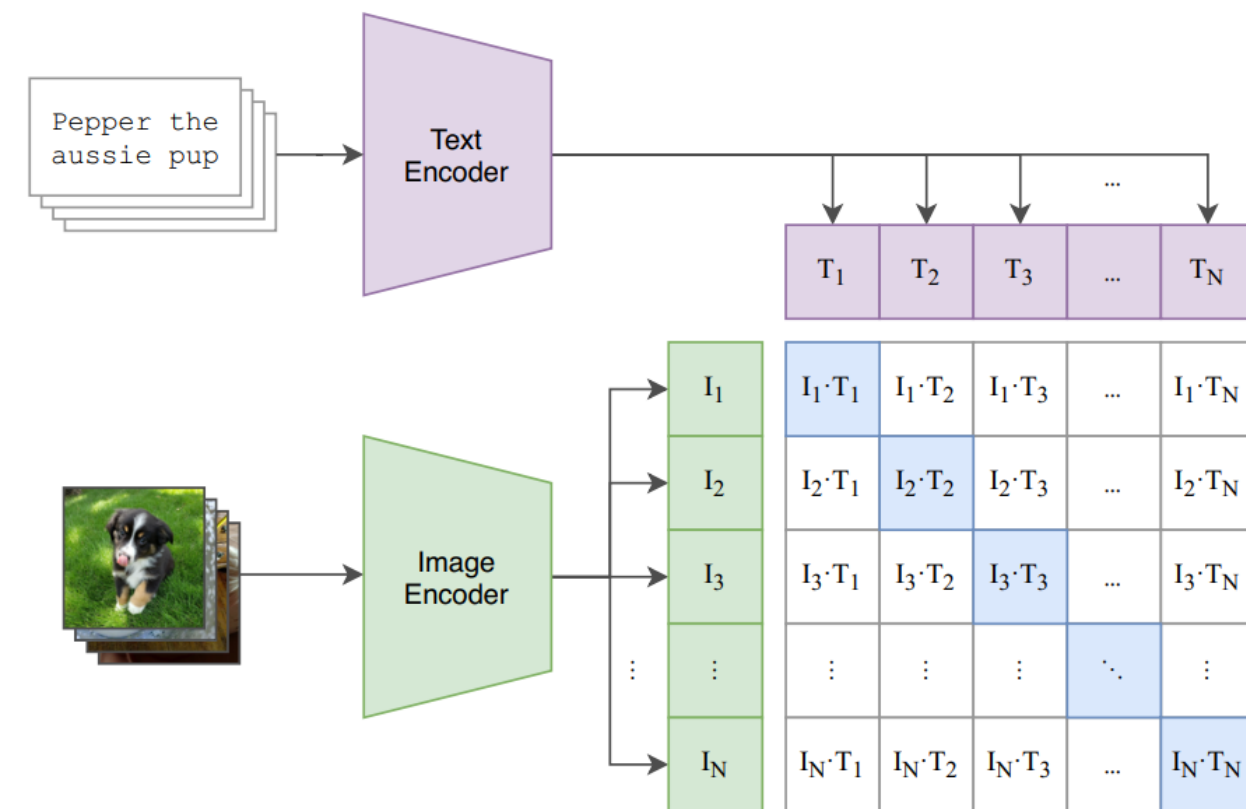
Why:

- Foundation models have a deep **understanding** of images and can be adapted to downstream tasks
- We want to avoid dataset **curation** and intensive **supervision**

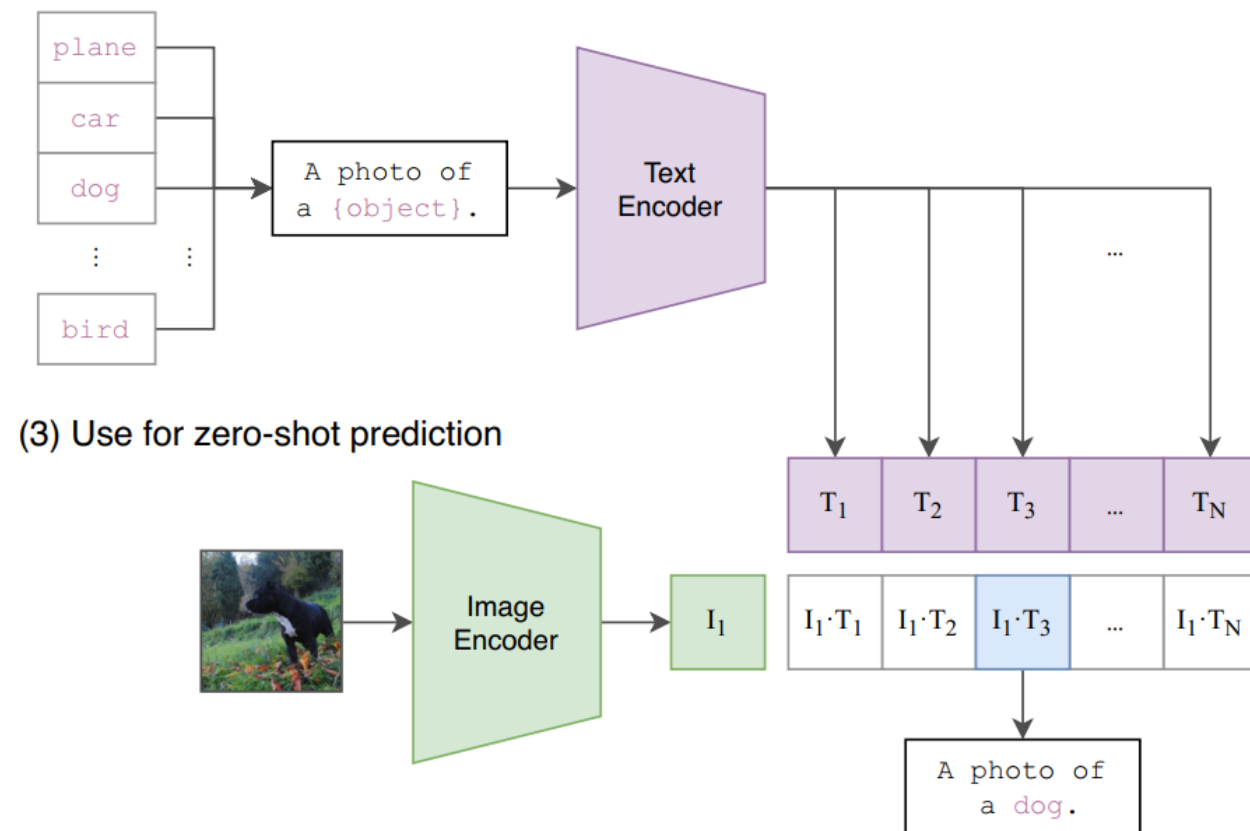
Abstract. Recent advances in deepfake generation have substantially increased the realism and diversity of manipulated content, rendering generalizable deepfake detection increasingly challenging. Most existing detectors perform well on known manipulation methods, however are challenged by distribution shifts, as they tend to overfit to dataset or manipulation-dependent artifacts. Vision foundation models such as CLIP provide rich semantic representations and have recently been adapted to deepfake detection using parameter-efficient fine-tuning (PEFT). However, when trained with a classical Real/Fake objective, PEFT-based adaptation often amplifies simplicity bias, causing the model to favor easily learnable shortcut features over manipulation-relevant forensic cues. In this work, we introduce a simplicity-bias-aware adaptation framework that enhances CLIP with discriminative local forensic cues while suppressing shortcut features. Our method integrates lightweight adapters with a feature-sieving mechanism, which identifies simplistic predictive cues and periodically erases them during training. This encourages the adapted model to learn richer, manipulation-agnostic forensic representations without constraining the pretrained backbone priors. We conduct extensive experiments on multiple deepfake benchmarks, demonstrating that our proposed method improves cross-dataset and cross-manipulation generalization, outperforming existing CLIP-based approaches. Our results highlight the importance of simplicity-bias mitigation for robust foundation-model adaptation in deepfake detection.

CLIP as a backbone for deepfake detection

(1) Contrastive pre-training



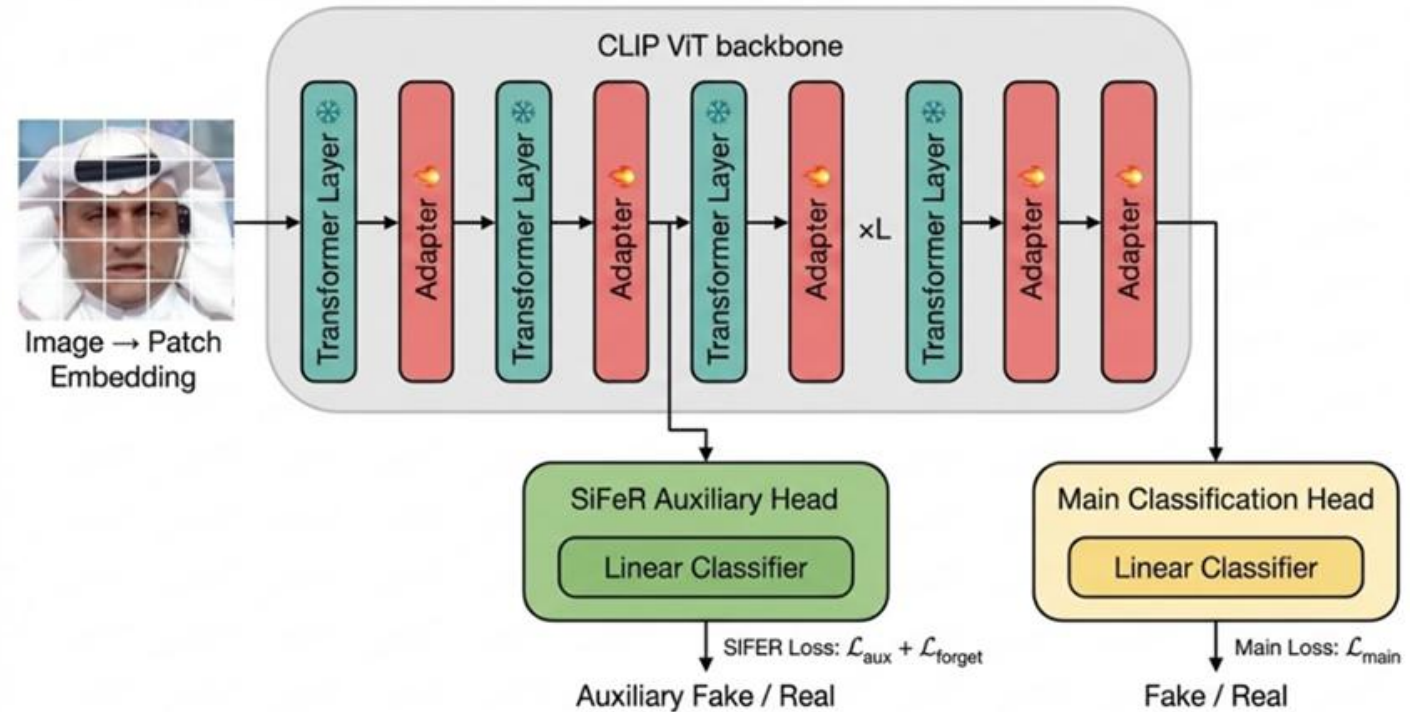
(2) Create dataset classifier from label text



● Our approach: Adapters + SiFeR

- **Adapter architecture:** linear projection + non-linearity + projection back
- Adapter layers are **trainable**, the backbone is frozen
- SiFeR auxiliary head:
 - Produce predictions $\hat{y}_{aux}$ from intermediate layers with a **simple linear head**
 - If the auxiliary head is “too good” => **bias** learning
 - Push $\hat{y}_{aux}$ towards a uniform distribution with a **forgetting loss**, every n steps
- **Advantage:** no need for bias annotation, dataset curation or supervision

$$A(F) = W_{up} \sigma(W_{down} F)$$



Yahchouchi et al. (submitted)

- Result: improved forensics learning



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● Take-away messages

- Deepfakes are becoming more and more **sophisticated** and **accessible**
- The spread of deepfakes will change our relationship to **media**
- Models that work well on face swaps mostly **fail** on face reenactment and vice versa
- Popular backbones struggle with forgeries located **in small areas**
- The **temporal** aspect is important, but hard to encode
- While deepfake detection works well on certain types of manipulation, we are **far from a one-fits-all model**

● Research directions

- Deeper understanding of deepfake clustering, how to set boundaries between deepfake families
- Work of the temporal dimension
- Mixture-of-Experts approach (ongoing)
- Explicability of models for a better problem framing



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Data Scientist



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Senior Researcher

● Method performance

- Cross-dataset

Detector	Backbone	CDF-v2	CDF-v3	DFD	DFDC	DFDCP
SBI [35]	EFNB4	93.2	–	82.7	72.4	–
UCF[42]	Xception	83.7	72	86.7	74.2	77
LSDA [39]	EFNB4	87.5	–	88.1	70.1	81.2
ProDet [5]	EFNB4	92.6	–	90.1	70.7	82.8
CDFA [24]	SwinTransformerV2	93.8	–	95.4	83.0	88.1
UDD [11]	CLIP ViT-B/16	93.1	–	95.5	81.2	88.1
P&P [44]	CLIP ViT-L/14	94.7	–	96.5	84.3	
ForADA [7]	CLIP ViT-L/14	95.7	75.6	97.2	87.2	92.9
DFD-FCG [14]	CLIP ViT-L/14	95	84	–	81.8	–
GenD [45]	CLIP ViT-L/14	96.0	85.9	97.0	87.1	–
EFFORT [40]	CLIP ViT-L/14	95.6	78.7	96.5	84.3	90.9
Ours	CLIP ViT-L/14	96.3	87.7	97.6	84.8	88.4

● Method performance

- Cross-manipulation

Method	UniFace	BleFace	MobSwap	FaceDan	FSGAN	InSwap	SimSwap	Avg.
F3Net [31]	80.9	80.8	86.7	71.7	84.5	75.7	67.4	78.2
SPSL [25]	74.7	74.8	88.5	66.6	81.2	64.3	66.5	73.8
RECCE [3]	89.8	83.2	92.5	84.8	94.9	84.8	76.8	86.7
SLADD [4]	87.8	88.2	95.4	82.5	94.3	87.9	79.4	87.9
SBI [35]	72.4	89.1	95.2	59.4	80.3	71.2	70.1	76.8
UCF [42]	83.7	83.1	82.7	73.1	86.2	93.7	80.9	83.3
IID [18]	83.9	78.9	88.8	84.4	92.7	78.9	64.4	81.7
LSDA [39]	87.2	87.5	93.0	72.1	93.9	85.5	79.3	85.5
ProDet [5]	90.8	92.9	97.5	74.7	92.8	83.7	84.4	88.1
CDFA [24]	76.2	75.6	82.3	80.3	94.2	77.2	75.7	80.2
EFFORT [40]	96.2	87.3	95.3	92.6	95.7	93.6	92.6	93.3
ForADA [†] [7]	95.1	87.9	97.9	94.6	97.5	94.8	91.3	94.2
Ours	98.0	94.4	98.3	97.4	98.8	97.1	95.0	97.0

- Out-of-distribution

Method	Wav2Lip	LIA	MCNet	FaceVid2Vid	DaNet	Avg.
UCF [42]	71.0	87.2	67.9	75.2	59.7	72.2
EFFORT [40]	64.6	79.6	64.0	71.5	62.0	68.3
ForADA [7]	57.8	79.5	54.2	62.8	50.1	60.88
Ours	81.3	93.4	78.9	87.6	73.3	82.9

- Teaser: easily trainable Mixture-of-Experts approach

